TruthLens

AI-Powered Document Forensics and Contextual Intelligence System

Observe → Correlate → Explain



DataOps Innovation Hackathon | Problem Statement 1

OCR reads documents. It does not verify them

1

Extracted ≠ authentic

OCR returns text, but cannot say whether a field was overwritten or a photo swapped.

2

Field checks miss context

Rules validate fields one by one. Conflicts only appear when documents are read together.

3

Manual review does not scale

Analysts compare ID, form and statements by eye, with no evidence trail.

Our goal: turn extraction into evidence-driven investigation.

End-to-end investigation pipeline



Every step writes to an audit log. Every field keeps its page, bounding box and confidence, so any finding traces back to the exact pixels it came from.

Architecture overview

Input layer	PDF, scans, images | upload UI | document IDs and hashes
Perception	OCR (printed + handwriting) | layout analysis | document classifier
Forensic engine	Pixel analysis | font and alignment checks | signature placement
Reasoning	Entity matching | rule + LLM contextual checks | cross-doc graph
Evidence store	Findings, regions, confidence, audit log | dossier PDF generator

Visual forensic analysis

1

Overwritten fields

Local noise and stroke-density breaks around edited text

2

Font and spacing drift

Glyph size, baseline and kerning compared to neighbours

3

Image tampering

Compression-level analysis and copy-move detection on photos

4

Signature checks

Placement, overlap with text, and similarity across documents

Suspicious regions are highlighted on the original page with a confidence score.

Contextual intelligence across documents

Field	Application form	ID proof	Bank statement	Status
Name	Rahul K. Sharma	Rahul Sharma	Rahul Sharma	Consistent
Date of birth	12-04-1991	12-04-1991	21-04-1991	Conflict
Address	Pune 411038	Pune 411038	Mumbai 400001	Conflict
Income	₹9,60,000	n/a	Avg ₹4,10,000	Check

Entity matching tolerates harmless variation (initials, spacing) and flags real conflicts. The LLM explains each conflict and suggests what to verify next.

Sample data is illustrative.

Three levels of finding, never "fraud"

Observed	A measurable fact: font size differs in one field, or a region has different noise.
Inconsistent	Two values conflict within or across documents. Needs explanation.
Potential manipulation	Several signals agree on one region. Still requires human review.

We never label a finding as confirmed fraud. A human makes the final call.

Dossier, traceability and audit trail

Investigation dossier (PDF)

- Document overview and extracted fields
- Contextual and cross-document findings
- Visual forensic findings with highlights
- Confidence and observation status
- Limitations and areas to verify

Click-to-trace

- Finding → source document, page, bounding box
- Hash-chained log of uploads, runs and findings
- Reviewer decisions recorded with timestamps

Technology stack and honest capability status

Component	Planned tools	Status
OCR / layout	PaddleOCR or Tesseract, LayoutLM	**Implemented**
Image forensics	OpenCV, ELA, copy-move	**Implemented**
Entity matching + LLM	Fuzzy matching, LLM reasoning	**Implemented**
Signature analysis	Placement + similarity	**Partial**
Handwriting, multilingual	Pretrained OCR models	**Planned**

Update the status column to match what your team actually builds before presenting.

Built for the judging criteria

20	15	15	15
Contextual intelligence	OCR + extraction	Visual forensics	Doc understanding

Live demo: upload genuine and tampered documents, see highlighted regions, open a cross-document conflict, trace it to the source pixel, download the dossier.

TruthLens: Observe → Correlate → Explain